

How to convert a LAN into WAN

BY

Prasad Nikam

Introduction

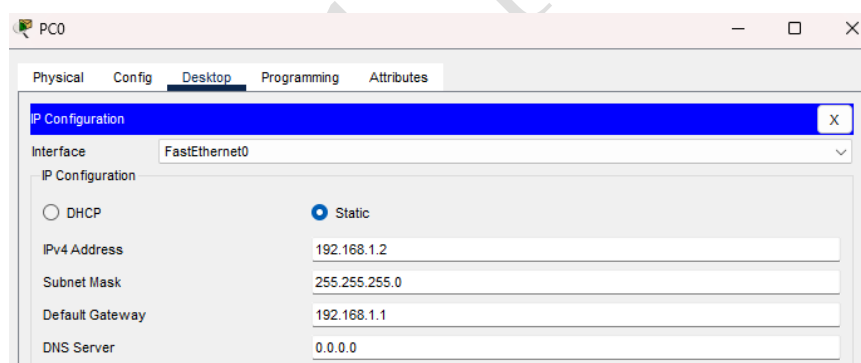
Computer networks enable multiple devices to communicate and share information with each other. Understanding how networks are created and how data travels between different networks is an important part of networking and cybersecurity. This practical report demonstrates the process of creating a basic Local Area Network (LAN) using PCs, switches, and routers, and then expanding the setup into a Wide Area Network (WAN) by connecting two separate LANs.

The LAN is first created by connecting multiple PCs to a central switch and assigning each device an appropriate IP address, subnet mask, and default gateway. This practical provides a basic understanding of IP addressing, subnetting, switches, routers, default gateways, static routing, and packet transmission. It also demonstrates how individual local networks can be interconnected to form a larger network environment.

Creating a LAN

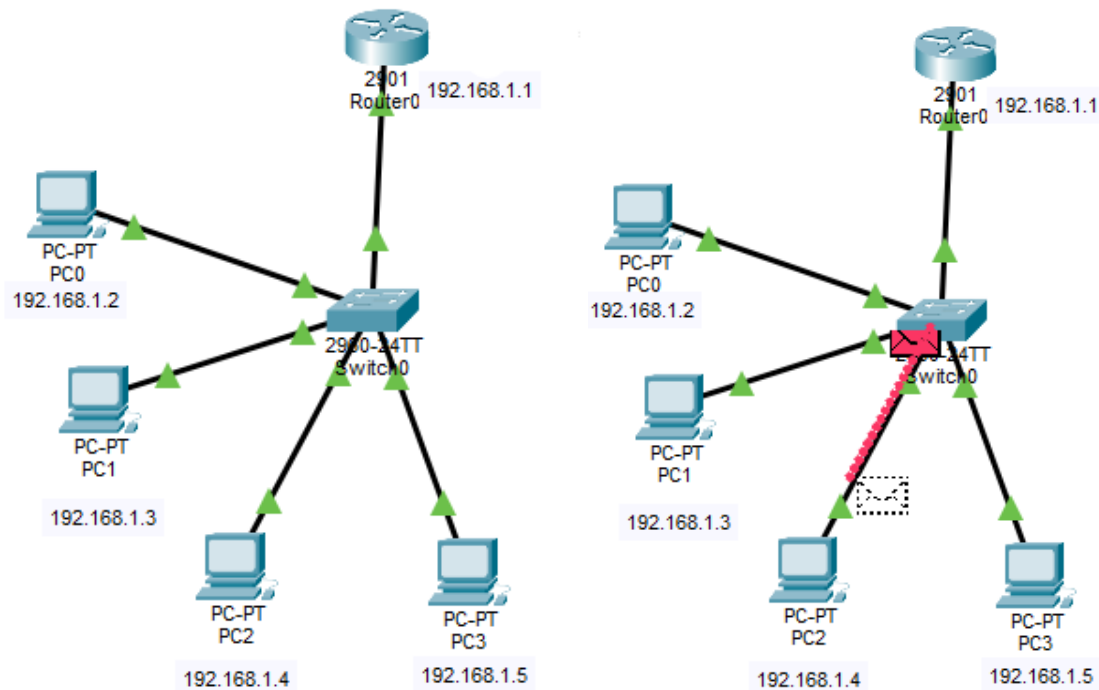
We start by making an environment of PC's connected to a switch as its central point then connected to a router which can be used later to expand our LAN. Connecting them with suitable cables. Then we give each PC its own IP address and subnet mask and default gateway (which is the IP we will give to router). For convention we can add a text near each PC indicating its IP address which will help us and others who review our work.

To give the PC its IP address we click on the PC then go to the desktop tab there are various icons, click on the IP configuration and simply enter the IP address, subnet mask, and default gateway.



After we give each PC its own IP they are then enabled to communicate with each other within the environment through the switch. We can check that through sending a message/packet from one PC to another.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)
	Successful	PC1	PC3	ICMP		0.000	N	1	(edit)
	Successful	PC3	PC0	ICMP		0.000	N	2	(edit)



The next step after verifying the connections and communication between PC's, we set up the router. Click on the router in the interface that popped up click on CLI and enter following commands:

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
exit

```

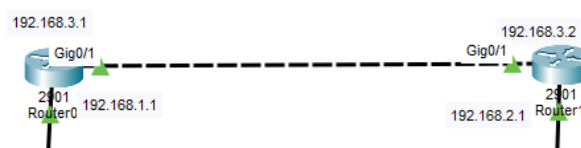
Now we verify that the router is working properly by sending a packet to the router form a PC.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC1	Router0	ICMP		0.000	N	0	(edit)

Expanding to WAN

WAN also known as Wide Area Network is basically a network consisting of 2 or more LAN's. So to create a WAN we will need two or more LAN's. Now, create another LAN following steps given previously (remember to give different IP series for this LAN). After we are done building both LAN's we will connect the routers of both LAN's and give them IP address, but we have already given the IP's so why? It's because the IP which was given previously was private IP for communicating with the network and now we are giving them the IP's for external communication i.e. public IP, which can be used to access or send information to another network. To do so we will open the CLI tab again and repeat the commands again but for another port which is connected to the other router.

How do you differentiate between ports, simply hover over the connection and between the routers and it will show you which ports is connected. As you can see in the picture below.



Now we will assign routers their public IP which will be of same series as you can see in the snapshot above.

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

Now the connection part is done but the packets will still not traverse between the two LAN's, because we have not given it a path to travel from one to another so that we will once again open the CLI tab in router and enter the following commands:

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

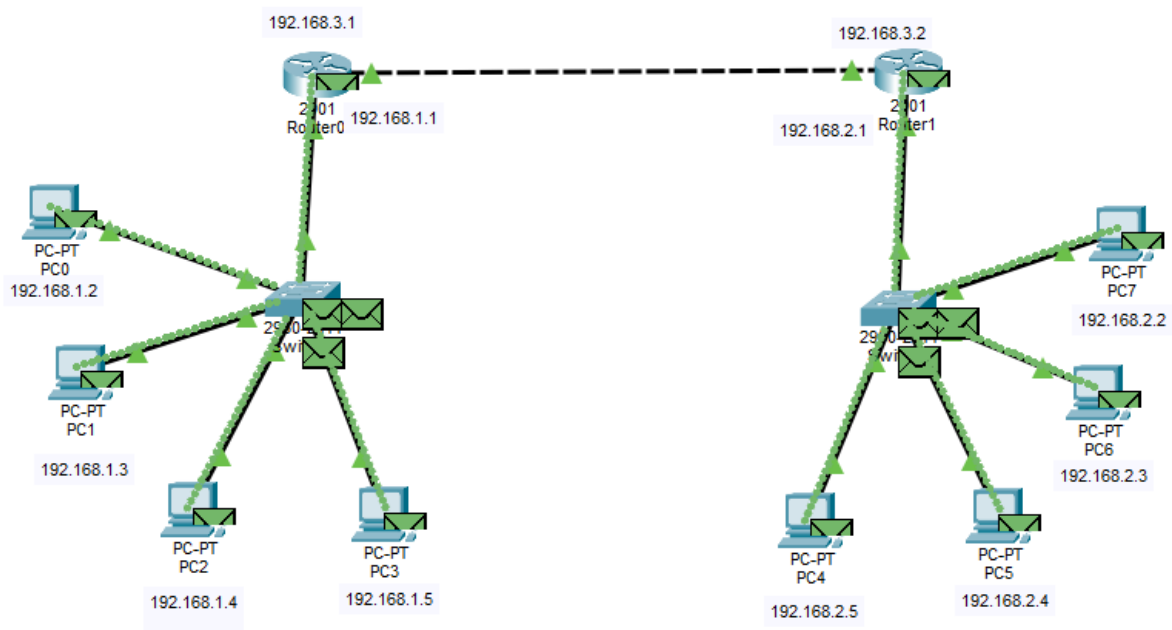
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
exit

```

The IP route command works as follows the first IP is the network series to where we want to share the packets followed by its subnet and then the path it's going to take to reach the destination i.e. the router which it will pass via.

After the both router are set up with the proper routes then the connection will be up and we will be able to send packets to and fro.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC1	PC6	ICMP		0.000	N	0	(edit)
	Successful	PC0	PC5	ICMP		0.000	N	1	(edit)
	Successful	PC7	PC2	ICMP		0.000	N	2	(edit)
	Successful	PC5	PC3	ICMP		0.000	N	3	(edit)



Conclusion

In this practical, a basic LAN was successfully created by connecting PCs through a switch and configuring their IP addresses and gateways. Two separate LANs were then connected through routers to form a WAN. Static routing was configured to enable communication between the networks. Overall, the practical provided hands-on understanding of IP addressing, switches, routers, and routing between different networks.